

Supporting Information

A synergy between site-specific and transient interactions drives the phase separation of a disordered, low complexity domain

Priyesh Mohanty^{1#}, Jayakrishna Shenoy^{2#}, Azamat Rizuan^{1#}, José F. Mercado Ortiz², Nicolas L. Fawzi^{2*}, and Jeetain Mittal^{1,3,4*}

¹Artie McFerrin Department of Chemical Engineering, Texas A&M University, College Station, TX 77843, United States

²Department of Molecular Biology, Cell Biology & Biochemistry, Brown University, Providence, RI 02912, United States

³Department of Chemistry, Texas A&M University, College Station, TX 77843, United States

⁴Interdisciplinary Graduate Program in Genetics and Genomics, Texas A&M University, College Station, TX 77843, United States

[#]These authors contributed equally

*corresponding authors; E-mail: nicolas_fawzi@brown.edu, jeetain@tamu.edu
Phone: (401) 863 5232, (979) 862-1920

Materials and Methods

All-atom PT-WTE simulation protocol

Force field choice and selection of temperature range. PT-WTE simulations of TDP-43 C-terminal domain (CTD, aa: 267-414) were performed using GROMACS 2018.8 (1) and PLUMED 2.5.4 (2). PT-WTE requires significantly fewer temperature replicas compared to conventional parallel tempering due to the amplification of potential energy fluctuations through periodic deposition of gaussian potentials (3). A total of 20 temperature replicas were generated based on a geometric progression that spanned a range from 300 to 475 K. The system topology was modeled using the AMBER99SBws-STQ force field with TIP4P/2005 water model (4-6). The force field files are available at https://bitbucket.org/jeetain/all-atom_ff_refinements/src/master/amber99sbws-stq.ff/.

Each replica was simulated for ~585 ns, resulting in an aggregate simulation time of ~12 μ s.

System topology and simulation setup. The initial structure for PT-WTE simulation was obtained from a CG simulation (1 μ s) of CTD (aa:267-414) using the HPS-Urry coarse-grained (CG) model (7, 8) with helical conformations in aa: 320-341 imposed using pseudo-dihedral potentials. The CG model was then converted into an atomistic model using MODELLER (9) and solvated using TIP4P/2005 (10) water molecules in an octahedral box ($l=12.5$ nm). 100 mM NaCl was added to mimic physiological salt concentration, and three additional Cl^- ions were added to achieve electroneutrality. Following energy minimization, temperature, and pressure equilibration of the system at 300 K and 1 bar, temperature replicas were prepared and equilibrated in the NVT ensemble without exchange for 10 ns at higher temperatures using the Langevin integrator ($t_i=1$ ps). Long-range electrostatic interactions were modeled using the Particle Mesh Ewald method (11, 12) with a real space distance cutoff of 0.9 nm. Following temperature equilibration of the replicas, production runs (timestep = 2 fs) were performed in the NVT ensemble with bias potentials deposited every 4 ps and exchanges attempted between adjacent replicas every 1 ps.

The initial height and width of the gaussian potential was set to 1.5 and 450.0 kJ mol⁻¹ respectively. Exchange probabilities between adjacent replicas varied from 20-40%.

Assessment of simulation convergence. The first 85 ns from the production replicas were discarded as equilibration time, during which the initial height of the gaussian potential (deposited every 4 ps) dropped below 0.25 kJ mol⁻¹, and sufficient overlap in the potential energy distributions (Fig. S2A) between adjacent temperature replicas was observed. Both 300 K and 346.5 K replicas underwent several round trips (Fig. S2B) through the entire temperature space on the ~100 ns timescale, which indicates efficient conformational exchange between replicas. The convergence of global properties was determined based on the radius of gyration (R_g) and end-to-end distance computed for the 300 K ensemble. The low errors associated with their probability distributions averaged over selected trajectory intervals indicate that the ensemble is well-converged (Fig. S2C).

Analysis of 300 K replica trajectory from all-atom PT-WTE simulation

Chemical shift calculation. Using the SPARTA+ algorithm (44), ¹³C chemical shifts ($\delta C_{\alpha/\beta}$, $\delta C'$) were computed from the 300 K replica trajectory every 200 ps and averaged over 500 ns (85-585 ns). Chemical shift Root Mean Square Deviation (RMSD) (Fig. S3A) for the CTD ensemble was computed with respect to experimental data reported in BMRB ID: 26823 using the equation:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\langle \delta \rangle_{\text{sim}} - \delta_{\text{exp}})^2}$$

Secondary chemical shifts ($\Delta \delta C_{\alpha/\beta}$) were computed with respect to random coil chemical shifts ($\delta C_{\alpha/\beta,rc}$) for the TDP-43 CTD sequence which were calculated using the Poulsen IDP/IUP random coil chemical shifts calculator at 300 K and pH 6.5 (13, 14) as:

$$\Delta\delta C_{\alpha} = \delta C_{\alpha,\text{sim}} - \delta C_{\alpha,\text{rc}}$$

$$\Delta\delta C_{\beta} = \delta C_{\beta,\text{sim}} - \delta C_{\beta,\text{rc}}$$

Experimental $^{13}\text{C}_{\alpha/\beta}$ chemical shifts were also referenced against the same random coil data set for comparison with $\Delta\delta C_{\alpha/\beta}$ from simulation (Fig. S3B). Standard errors for secondary chemical shifts were calculated over five, non-overlapping blocks of data (100 ns each).

Intramolecular contact analysis. In the 2D residue index contact maps (Fig. 2B), N_{contact} is the average number of pairwise contacts per frame between residue i and j calculated over N frames

as: $\frac{\sum_{k=1}^N n_{ij}}{N}$. $n_{ij} = 1$ if the contact distance criteria for residues i and j is satisfied and $n_{ij} = 0$ if not.

Residues i and j are in contact in a specific simulation frame if the distance between any pair of heavy atoms belonging to amino acids with sequence separation ($|i-j|$) greater than 3 was below 0.6 nm. For assessing the correlation between residue-level contacts based on interaction modes (Fig. 2B) and their corresponding interaction energies, per-residue interaction energies were computed using gRINN (15) (<https://bitbucket.org/onursercinoglu/grinn/src/master/>).

Coarse-grained simulation protocol and analysis

Simulation protocol. Phase coexistence simulations of TDP-43 CTD wild type and mutants were conducted using the HOOMD-Blue 2.9.3 software package (16), according to the protocol (https://bitbucket.org/jeetain/hoomd_slab_builder/src/master/) described in our previous studies (17). Residue beads from the conserved helix (aa:320-341) domain were kept rigid during the CG simulations to mimic the α -helical structure present in CR, using the `hoomd.md.constrain.rigid` function. The initial slab configuration (15 nm $\times$ 15nm $\times$ 150 nm) was prepared from 100 chains of the TDP-43 CTD sequence based on the HPS-Urry model (8) and simulated for 5 μ s in the

NVT ensemble at 300 K using a time step = 10 fs. The Langevin thermostat was used for temperature control, with the damping factor (τ) set to 1000 ps.

The CG single chain simulation of wild type TDP-43 CTD was conducted at 300 K based on the HPS-Urry model (8). The simulation utilized the LAMMPS software package (18) and was performed in the cubic box (50 nm $\times$ 50 nm $\times$ 50 nm). Temperature was maintained using a Langevin thermostat, with the damping factor (τ) set to 1000 fs.

Trajectory Analysis. Radius of gyration, coexistence densities and pairwise contacts were calculated using the protocol mentioned in our previous work (7, 17). During all analysis, the first 1 μ s was treated as an equilibration period and omitted for subsequent calculations. For intermolecular contact analysis, N_{contact} was computed via summation of contacts for each residue pair - i and j, with contributions arising from pairs of chains satisfying the distance criteria for contact formation. Two residues i and j were considered to form a contact if the distance between them was less than $1.5\sigma_{ij}$, where σ_{ij} is the arithmetic average of the diameters of residues i and j. Contacts within the condensed phase are normalized by the number of chains (n=100), to obtain values that are independent of the system size (19). In the accompanying 1D residue index contact map, N_{contact} represents the summation over average pairwise contacts at each residue position.

Expression and purification of recombinant proteins

Wild-type human TDP-43 CTD and mutant variants of TDP-43 CTD were expressed from a codon-optimized sequences in the pJ411 bacterial expression vector. These proteins were expressed in BL21 Star (DE3) Escherichia coli cells (Life Technologies). The expression and purifications were either in LB or M9 minimal media supplemented with $^{15}\text{NH}_4\text{Cl}$ with slight modifications as previously described (20, 21). Briefly, the bacterial cultures were induced at OD

0.8 with 1 mM IPTG and grown for 4h at 37°C and 200 rpm. The cells were harvested by centrifugation (6000 rpm, 15 minutes, 4°C). Cell pellets thus obtained were resuspended in 20 mL buffer (20 mM Tris, 500 mM NaCl, 10 mM imidazole, pH 8.0), lysed with an ultrasonic cell disrupter, and centrifuged (20,000 g, 1 hr, 4°C). The insoluble material with the inclusion bodies was resuspended in 40 ml solubilizing buffer (8M urea, 20 mM Tris, 500 mM NaCl, 10 mM imidazole, pH 8.0), and centrifuged (25,000 rpm, 1hr, 19°C). The supernatant was filtered by 0.22 µm syringe filter, and the protein was purified using a 5 mL Histrap HP column with a gradient of 10 to 500 mM imidazole added to the insoluble buffer. The pure protein fractions were desalted using a HiPrep 26/60 Desalting Column into TEV cleavage buffer (20 mM Tris, 500 mM GdnHCl, pH 8.0) for TEV reaction overnight at room temperature. After cleavage, the solid urea was added to the solution to approximately 8 M urea and the resolubilized protein was applied to the Histrap HP column to remove histidine-tag and histidine-tagged TEV protease. Cleaved protein fractions were concentrated and buffer-exchanged into storage buffer (20 mM MES, 8 M urea, pH 6.1), flash-frozen, and stored at -80°C.

Microscopy

According to the manufacturer's instructions, using 0.5 mL Zeba spin desalting columns (Thermo Scientific), the protein samples were desalted to MES buffer (20 mM MES, pH 6.1), and diluted to 40 µM, supplemented with varying concentrations of 0, 150, and 300 mM NaCl. Samples were gently mixed and monitored for phase separation in DIC micrographs with a Zeiss Axiovert 200M microscope with a 40x objective after spotting 10 µL of sample onto the coverslip. The images were processed with ImageJ.

Determination of saturation concentration for quantification of *in vitro* phase separation

For a quantitative analysis of CTD mutant phase separation, we conducted assays to determine the saturation concentration wherein the protein concentration in the supernatant was measured after centrifugation of samples with increasing salt concentration to induce phase separation. During centrifugation, the micrometer-sized droplets sediment account for protein in the phase-separated state, while the sample remaining in the supernatant accounts for the protein content in the dispersed phase. The amount of protein remaining in the supernatant represents the saturation concentration (the concentration above which the protein phase separates) and gradually reduces with increasing salt concentrations and is correlated with phase separation propensity. The desalted protein samples were diluted to 80 μM or 120 μM using the MES buffer. Phase separation was initiated by adding an equal volume of NaCl stock solution prepared in MES buffer to obtain final salt concentrations of 0, 75, 150, 300, and 500 mM NaCl. Then, the samples were gently mixed and centrifuged for 10 minutes at 12,000 rpm. Later, the protein concentration in the supernatant was measured using the ($A_{280}/E_{280} = 18,350 \text{ M}^{-1} \text{ cm}^{-1}$) with a Nanodrop 2000c. All data points were measured in triplicate.

NMR spectroscopy

All NMR experiments were performed Bruker Avance 850 MHz or 600 MHz ^1H Larmor frequency spectrometers with HCN TCI z-gradient cryoprobes at 298K. The NMR samples were prepared in 20 mM MES Buffer, pH 6.1, supplemented with 10% D_2O . Backbone assignments for the CTD variant were obtained by standard triple resonance experiments (HNCACB, HNCA, CBCACONH, and HNN (HNCANNH)). The NMR data were processed with Bruker Topspin and analyzed using CCPNMR (22). Experimental NMR parameters summary can be found from our previous works (20, 21).

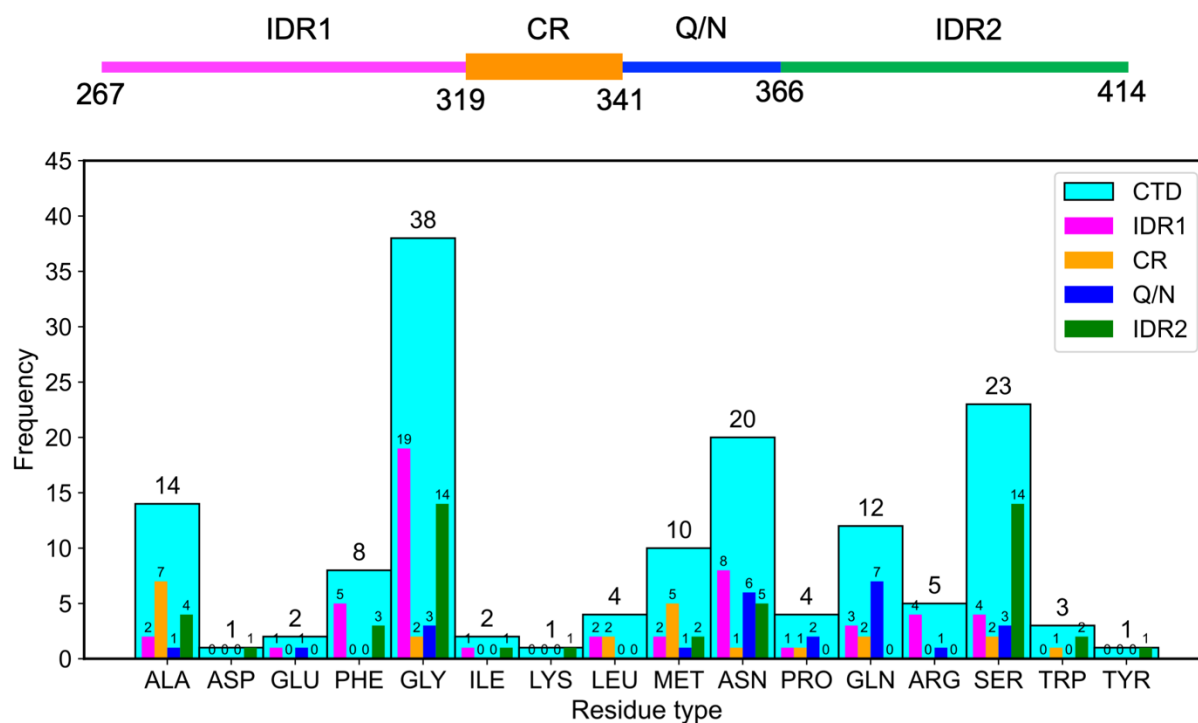


Figure S1. Sequence composition of TDP-43 C-terminal domain (CTD) and its subregions.

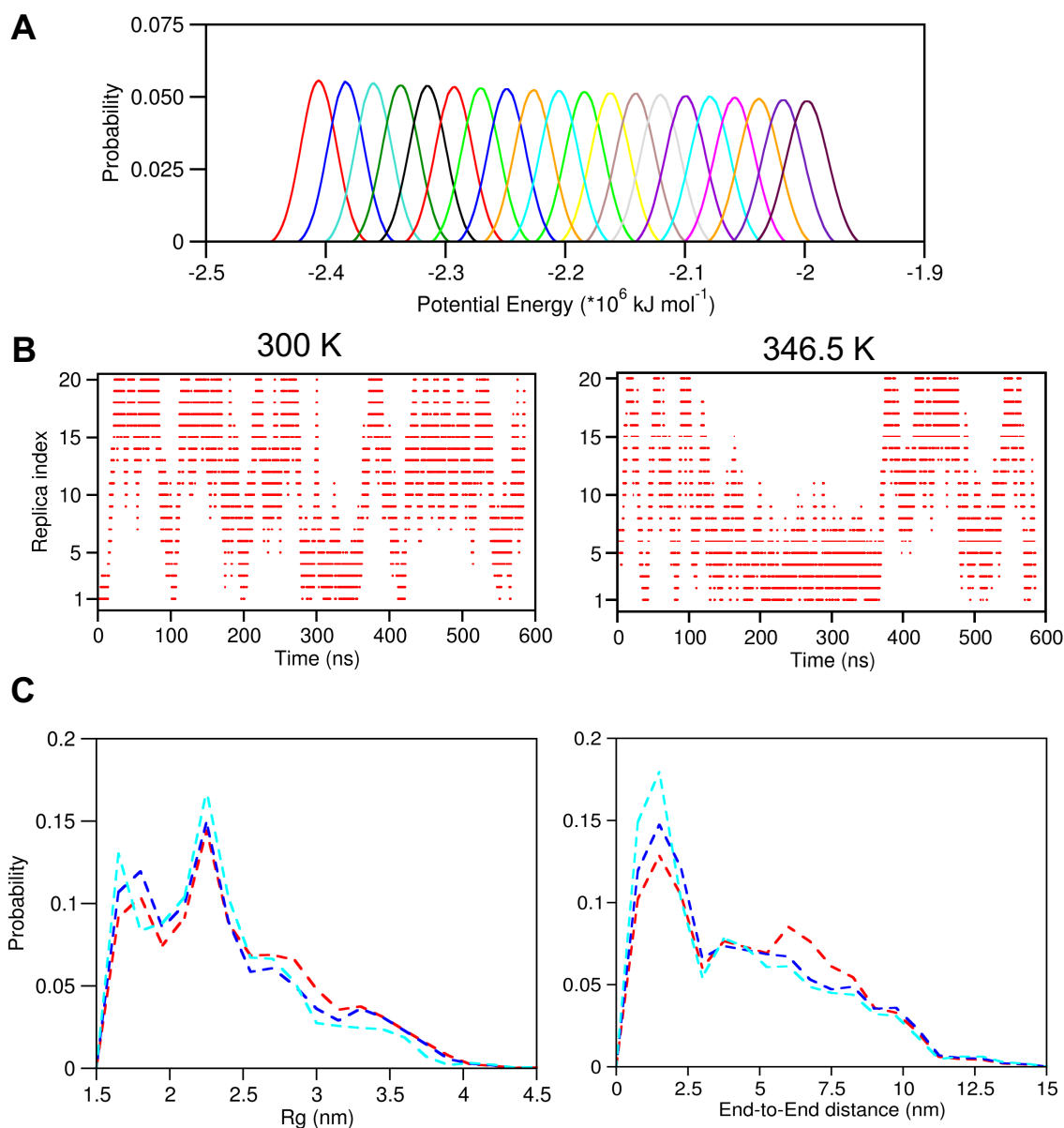


Figure S2. Assessing convergence of the all-atom PT-WTE ensemble. **A.** Potential energy distributions for all 20 replicas. Distributions were calculated over ~ 585 ns per replica and show good overlap between adjacent replicas in temperature space. **B.** Movement of 300 K and 346.5 K replicas through temperature space because of conformational exchange between adjacent replicas. **C.** Radius of gyration (left) and end-to-end distance probability (right) distributions (dashed lines) of the 300 K replica calculated over 585 (red), 500 (blue) and 400 ns (cyan) largely overlap indicating convergence of these quantities across trajectory intervals.

A

Chemical shift	RMSD (ppm)
C_α	0.49
C_β	0.46
C'	0.53

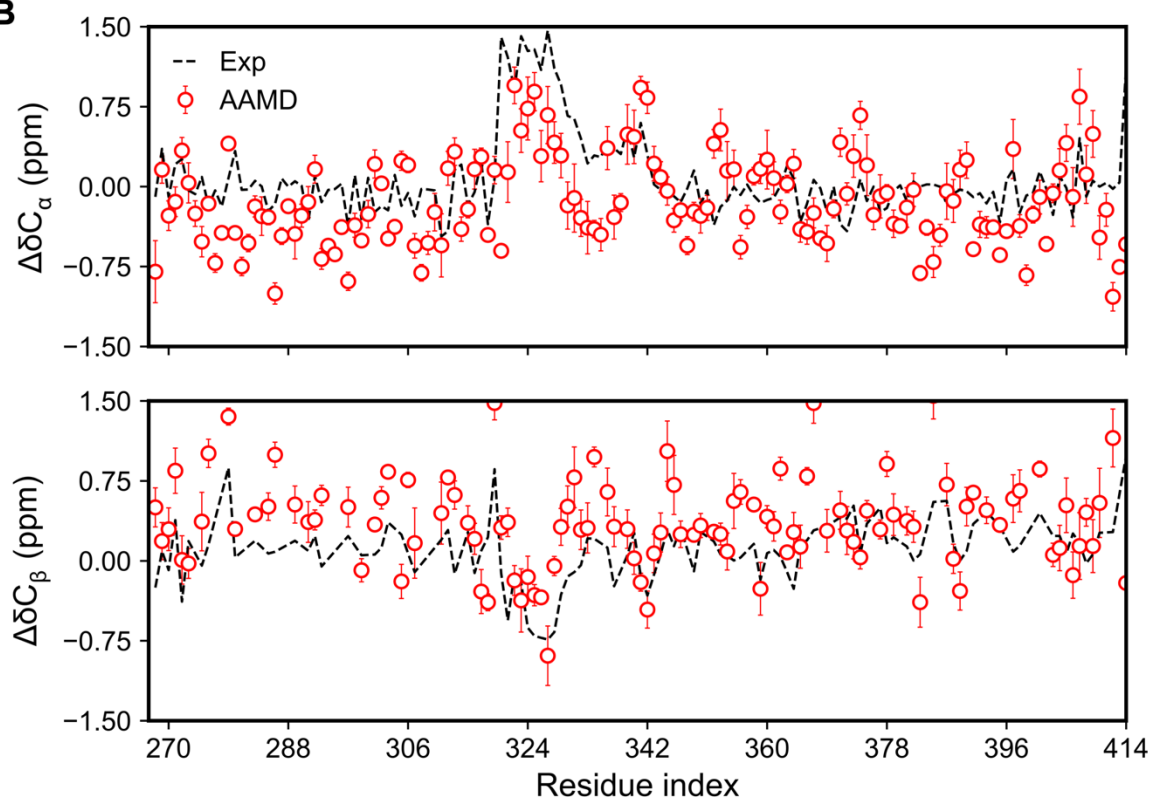
B

Figure S3. Validation of AA ensemble of CTD at 300 K based on NMR chemical shifts (δ). **A.** Carbon chemical shift RMSDs between the 300K replica and experiment. **B.** Comparison of per-residue secondary chemical shifts ($\Delta\delta C_\alpha$, $\Delta\delta C_\beta$) between 300 K simulation replica and experiment.

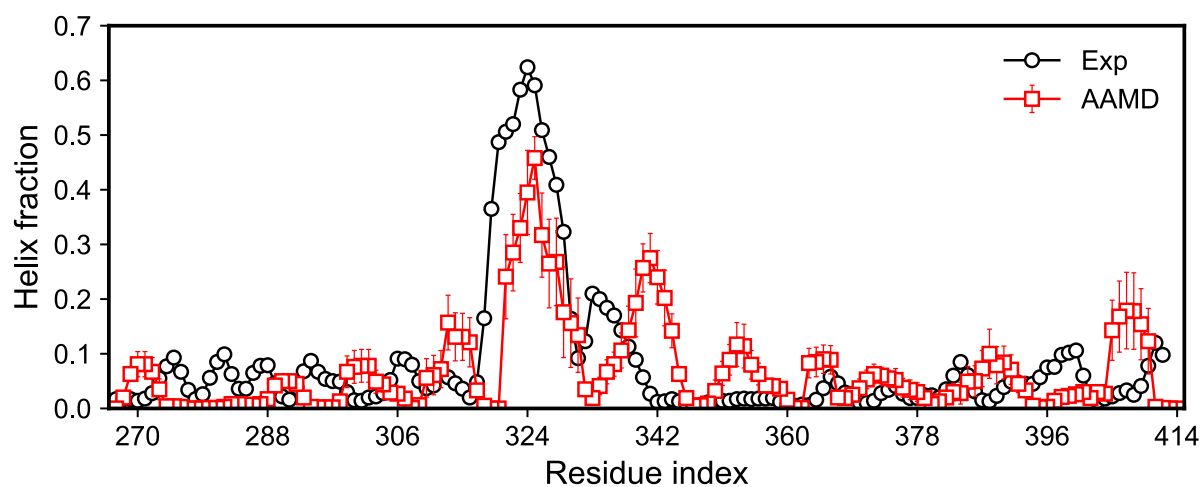


Figure S4. Comparison of per-residue helical fractions from single chain AA simulation of CTD (300 K replica trajectory) using the DSSP algorithm (23) with NMR-based fractions calculated using δ 2D (24).

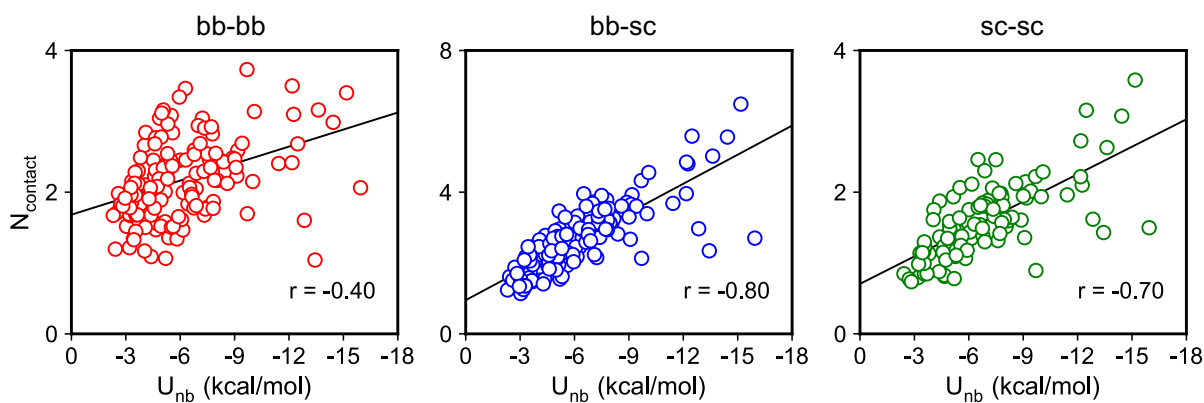
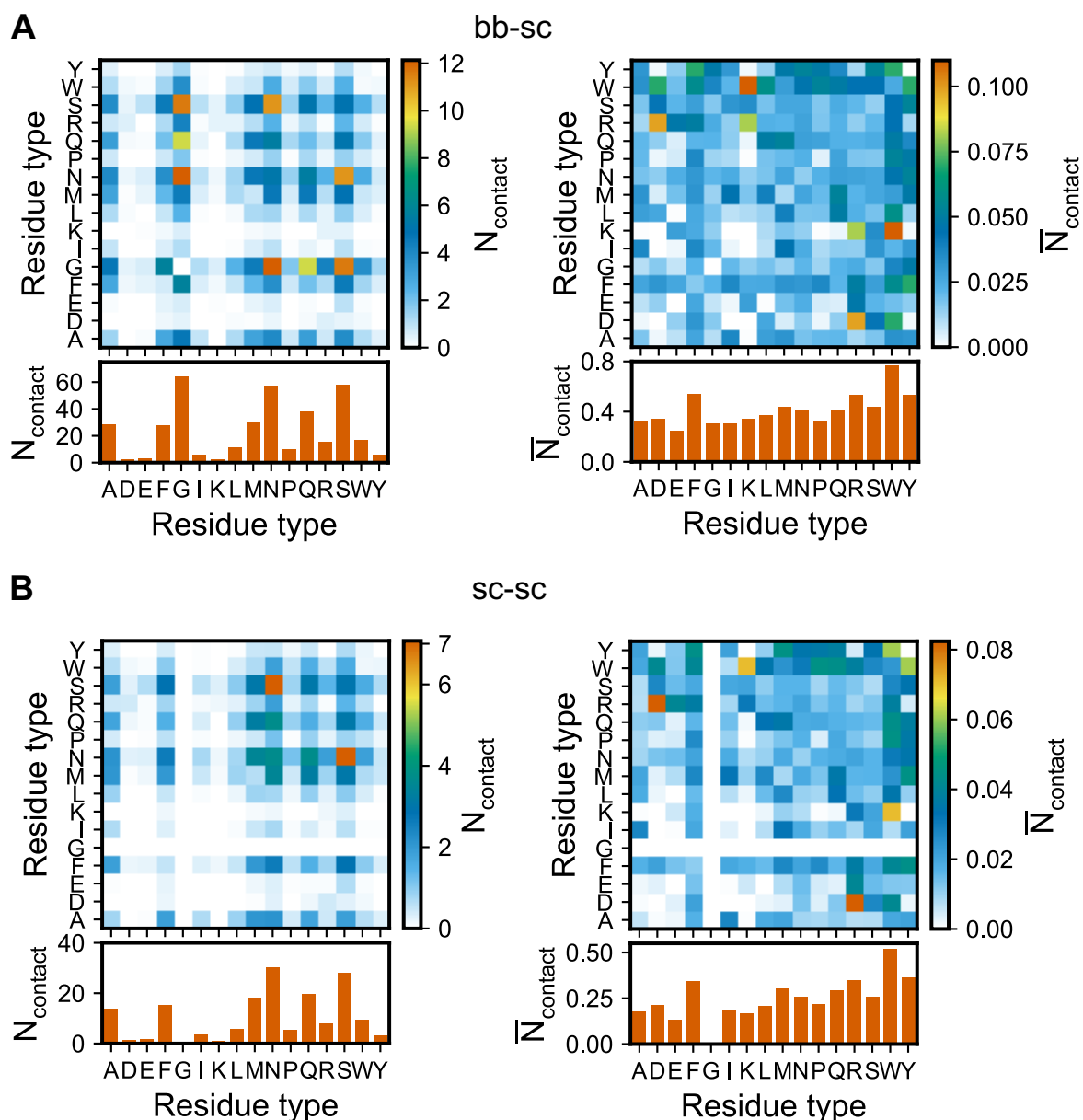


Figure S5. Correlation between per-residue intramolecular contacts (N_{contact}) and non-bonded interaction energies (U_{nb} , coulombic + van der Waals) for neutral residues calculated from single-chain AA simulations (300 K replica trajectory). N_{contact} is calculated via summation over all pairwise contacts shown in the 2-D contact maps in Fig. 2B for each residue position. The correlation was computed for three different interaction modes - backbone-backbone (bb-bb), backbone-sidechain (bb-sc), and sidechain-sidechain (sc-sc).



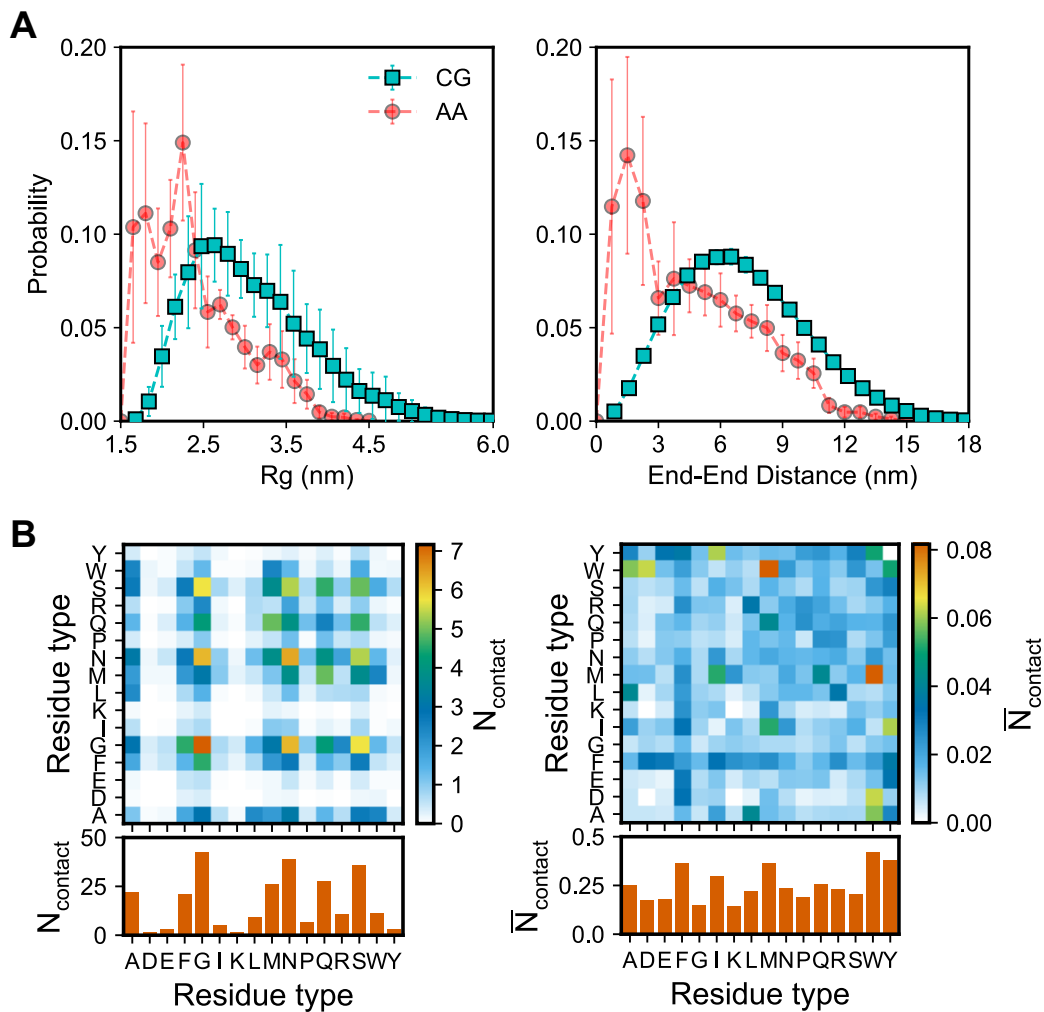


Figure S7. A. Radius of gyration and end-end distance probability distributions of single chain TDP-43 CTD from CG simulations calculated over 5 μs . The first 1 μs was excluded as equilibration time. Standard errors are estimated using block averages with four blocks. **B.** Intramolecular pairwise interactions between all residue types occurring in single-chain CTD from CG single-chain simulations. N_{contact} corresponds to absolute number of pairwise interactions while $\bar{N}_{\text{contact}}$ is normalized by the total count of each residue type occurring in the TDP-43 CTD sequence. The bottom panels for each contact map indicate the 1-D summation (along y-axis) over all pairwise $N/\bar{N}_{\text{contact}}$ values shown in the corresponding 2-D contact maps for each residue type.

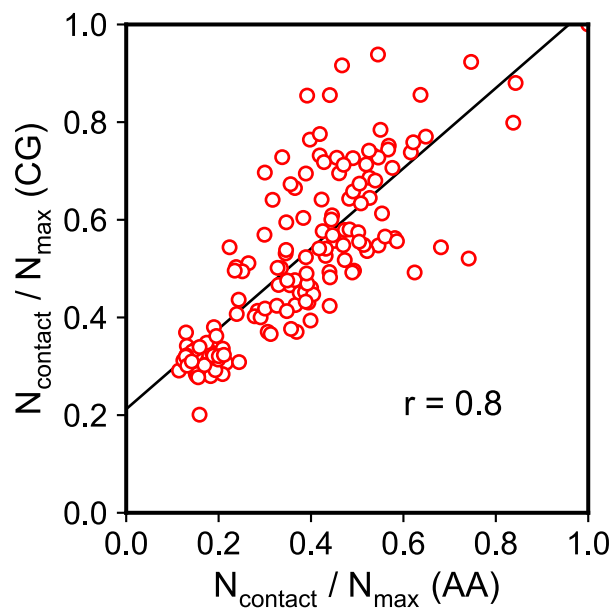


Figure S8. The correlation between average number of per-residue intramolecular and intermolecular contacts calculated from all-atom single-chain simulations (AA), and coarse-grained (CG) phase coexistence simulations, respectively. The probabilities were normalized by the maximum value observed for a given residue in each case.

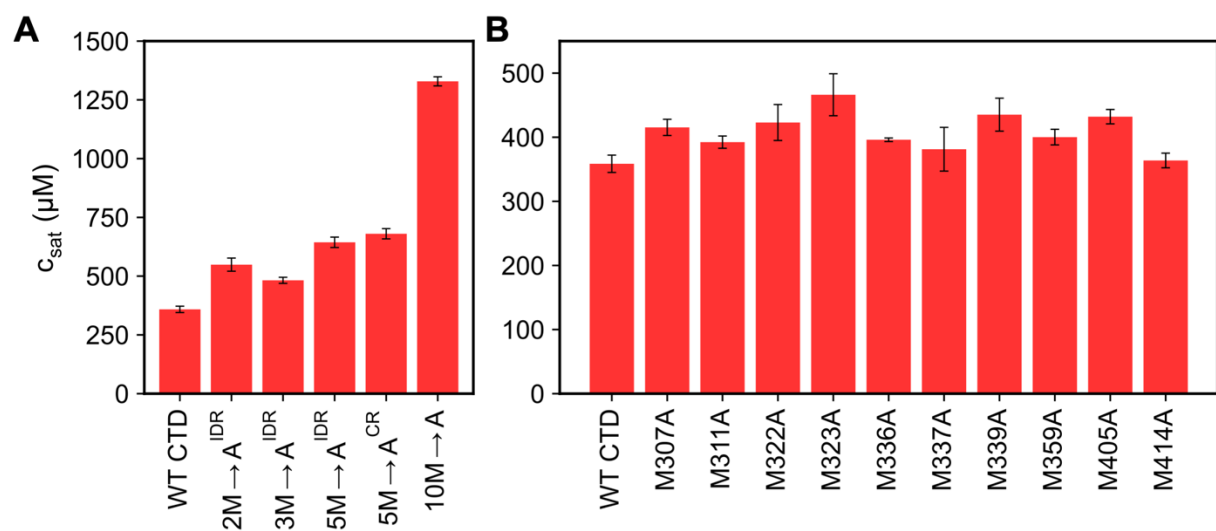


Figure S9. A. Saturation concentration (c_{sat}) of bulk and, **B.** positional methionine mutations for TDP-43 CTD computed from CG phase coexistence simulations.

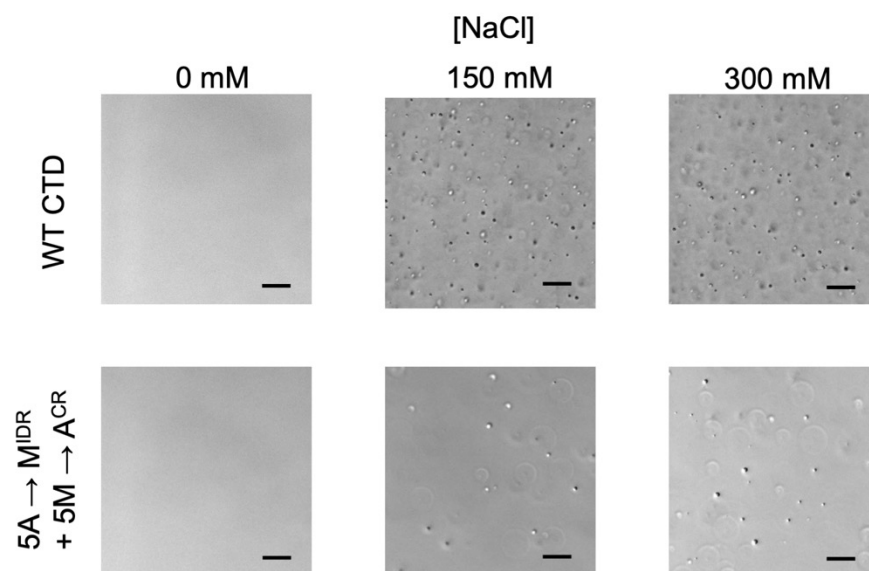


Figure S10. DIC micrographs for wild type (WT) TDP-43 CTD and A→M^{IDR} + 5M→A^{CR} variant at different salt concentrations and 80 μ M protein concentration. Scale bar, 20 μ m.

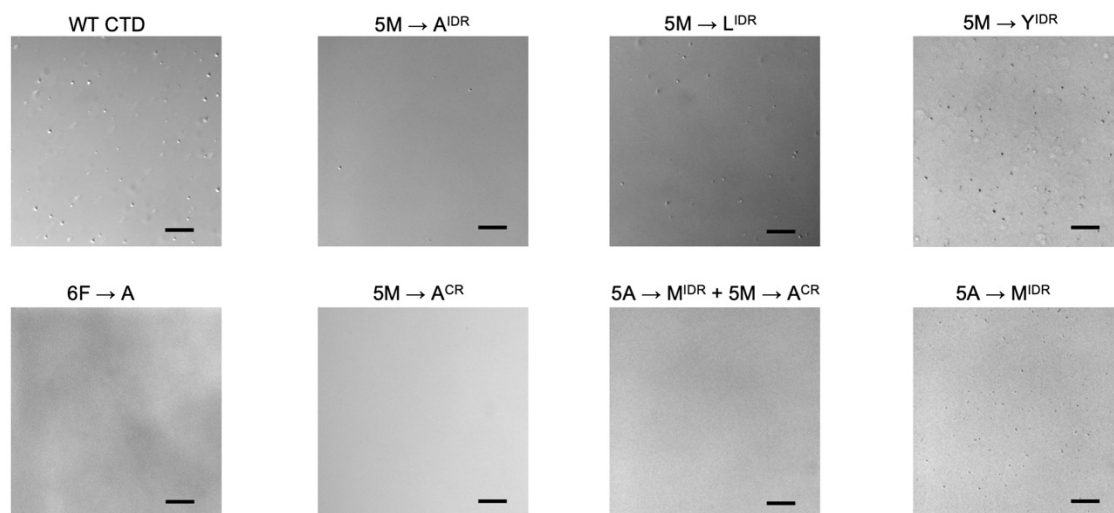


Figure S11. DIC micrographs for wild type (WT) TDP-43 CTD and its designed variants at 150 mM salt and 40 μ M protein concentration. Scale bar, 20 μ m.



Figure S12. The sequence logo of TDP-43 CTD is based on the sequence alignment of 93 vertebrate homologs (25). Sequence alignment was carried out using MEGA11 based on the MUSCLE option (26). Sequence logos were produced by using WebLogo 3 (27). The gap between M307 and G308 comes from *Xenopus laevis* CTD homolog that has additional residues (GSGGGG) between M307 and G308 present in homo sapiens CTD.

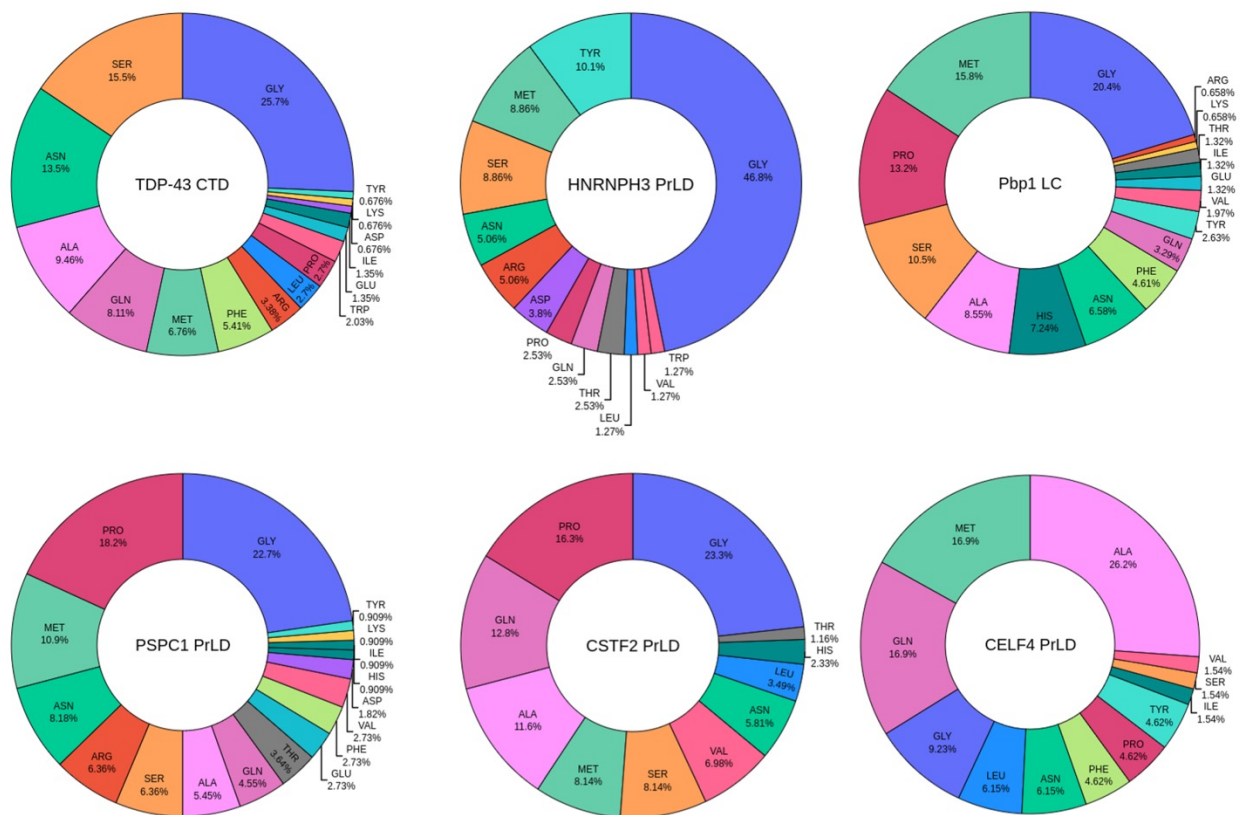


Figure S13. Residue composition of human RNA-binding proteins with prion-like domains (PrLDs) that have significant amount of methionine composition.

Table S1. TDP-43 CTD mutants were tested in CG condensed phase simulations and *in vitro* assays.

Mutant	Residues
6F→A	F276A, F283A, F289A, F367A, F397A, F401A
5M→A ^{CR}	M322A, M323A, M336A, M337A, M339A
5M→X ^{IDR}	M307X, M311X, M359X, M405X, M414X, where X = A, L, Y
5A→M ^{IDR} + 5M→A ^{CR}	A297M, A367M, A381M, A388M, A391M M322A, M323A, M336A, M337A, M339A
5A→M ^{IDR}	A297M, A367M, A381M, A388M, A391M

Table S2: Methionine composition analysis of the 29 human RNA-binding proteins with prion-like domains (PrLDs) (28), including ataxin-2, show that one-third of PrLDs is comprised of more than 4% MET.

PrLD name	% MET	# MET
Pbp1 LC (ataxin-2)	15.79	24
CELF4 PrLD	16.92	11
PSPC1 PrLD	10.91	12
HNRNPH3 PrLD	8.86	7
CSTF2 PrLD	8.14	7
TDP-43 CTD	6.76	10
HNRNPH1 PrLD	5.88	4
HNRNPH2 PrLD	5.88	4
RBM33 PrLD	4.27	5
TIA1 PrLD	4.21	4

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